Alla scoperta del Vector data type

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About me

BI Architect Lucient ITALIA
25 years burning on “Data Realm”
Giving my little contribution to the community
Power BI User Group Italy co-Founder & Administrator

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- AI everywhere
- A lot of different AI
- AI on my own data?
- Vector is the key
- Vector & Embeddings
- Vector data type
- Vector Functions

OPTIMIZATION
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SIMULATION

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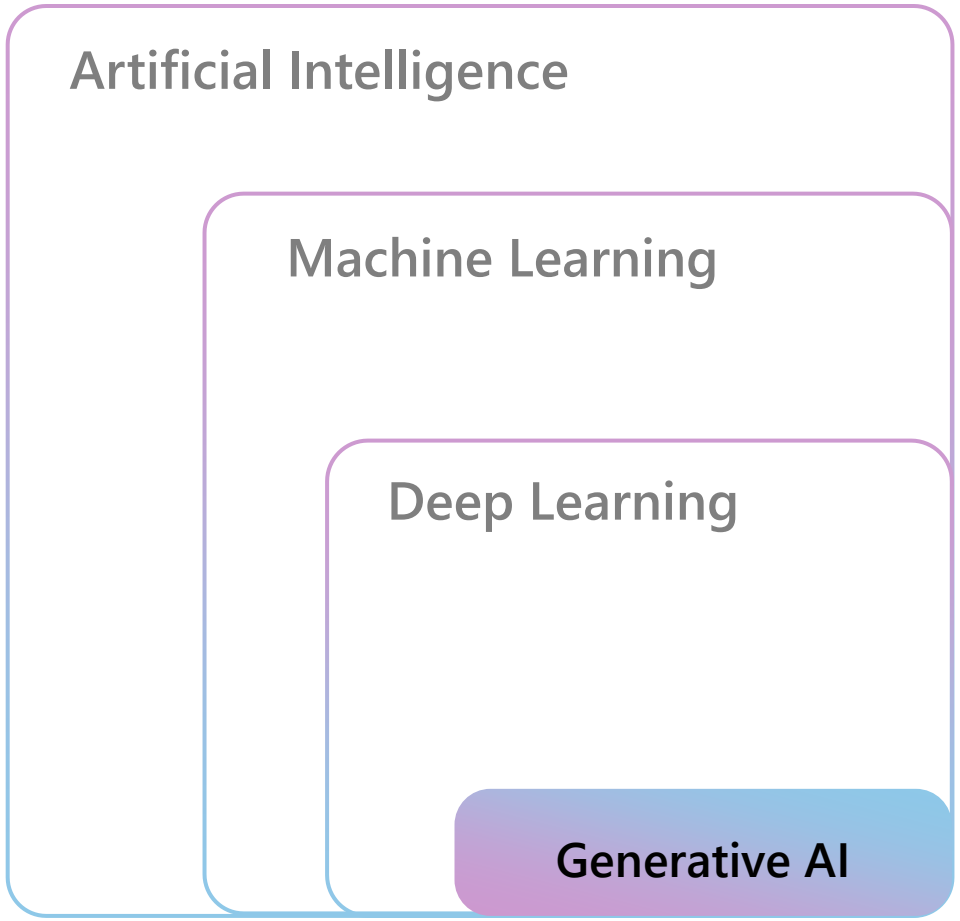
This is a new moment for AI



1950s

Artificial Intelligence

the field of computer science that seeks to create intelligent machines that can replicate or exceed human intelligence.

1959

Machine Learning

subset of AI that enables machines to learn from existing data and improve upon that data to make decisions or predictions.

2017

Deep Learning

a machine learning technique in which layers of neural networks are used to process data and make decisions.

2021

Generative AI

create new written, visual, and auditory content given prompts or existing data.

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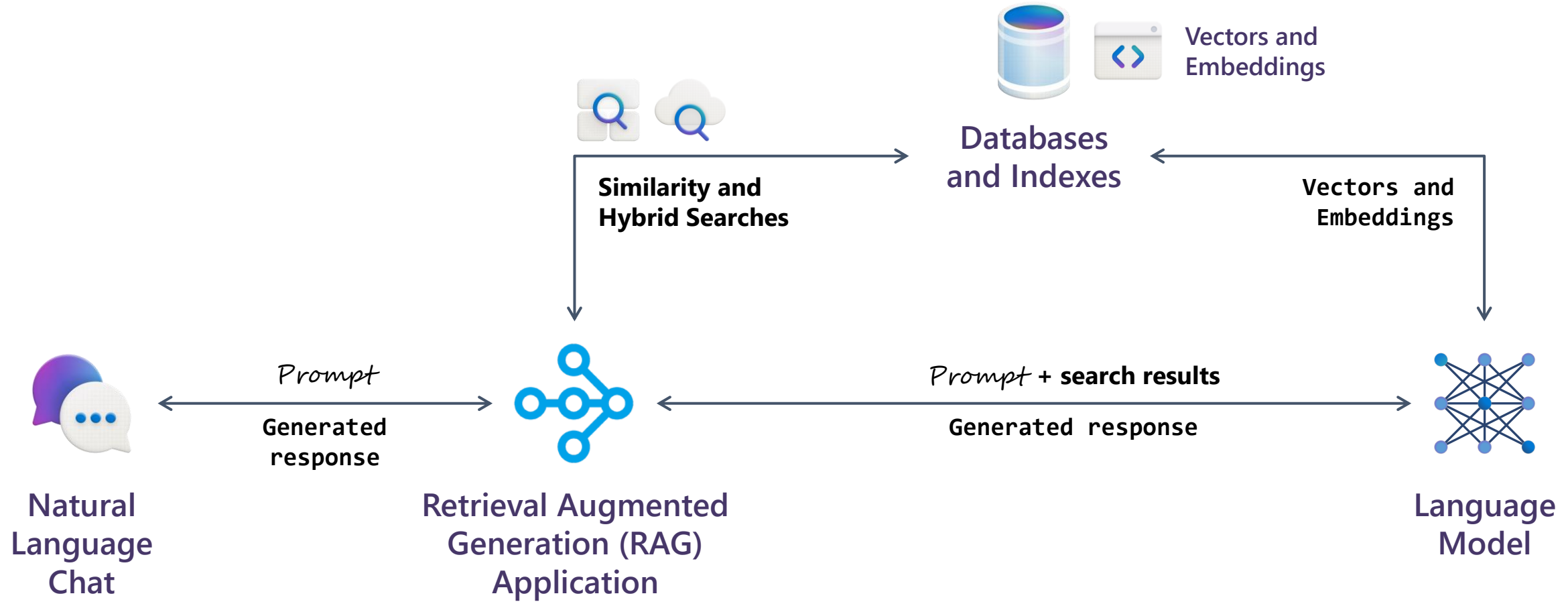
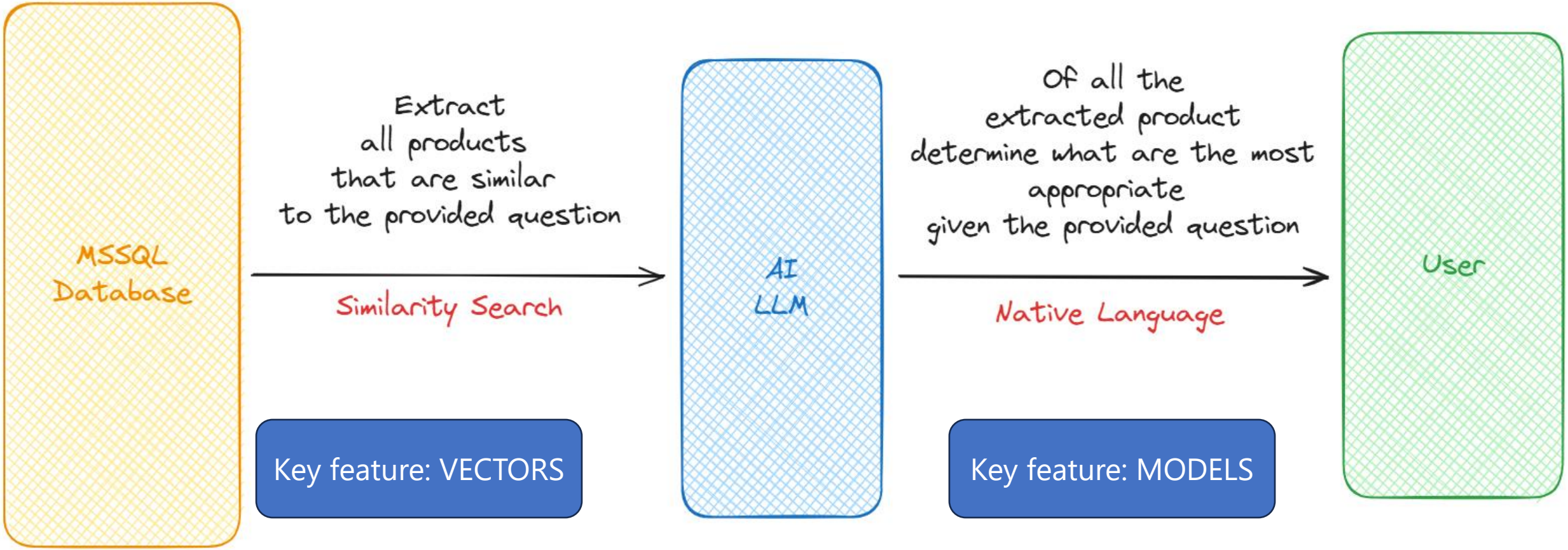
Building Generative AI applications with databases



RAG (Retrieval Augmented Generation) Pattern



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Get smarter with your data

Precise

Tailored

Faster

Intelligent

Interactive

Prompt sessions using natural language

Images created by Microsoft Copilot

Generative AI on my **ON PREMISES** data?

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SI PUÒ FARE!

Feature Vector

Ordered array of numbers typically created by a human to train a model

[*Height, Weight, Age, Fur Length, Energy Level*]

Embedding

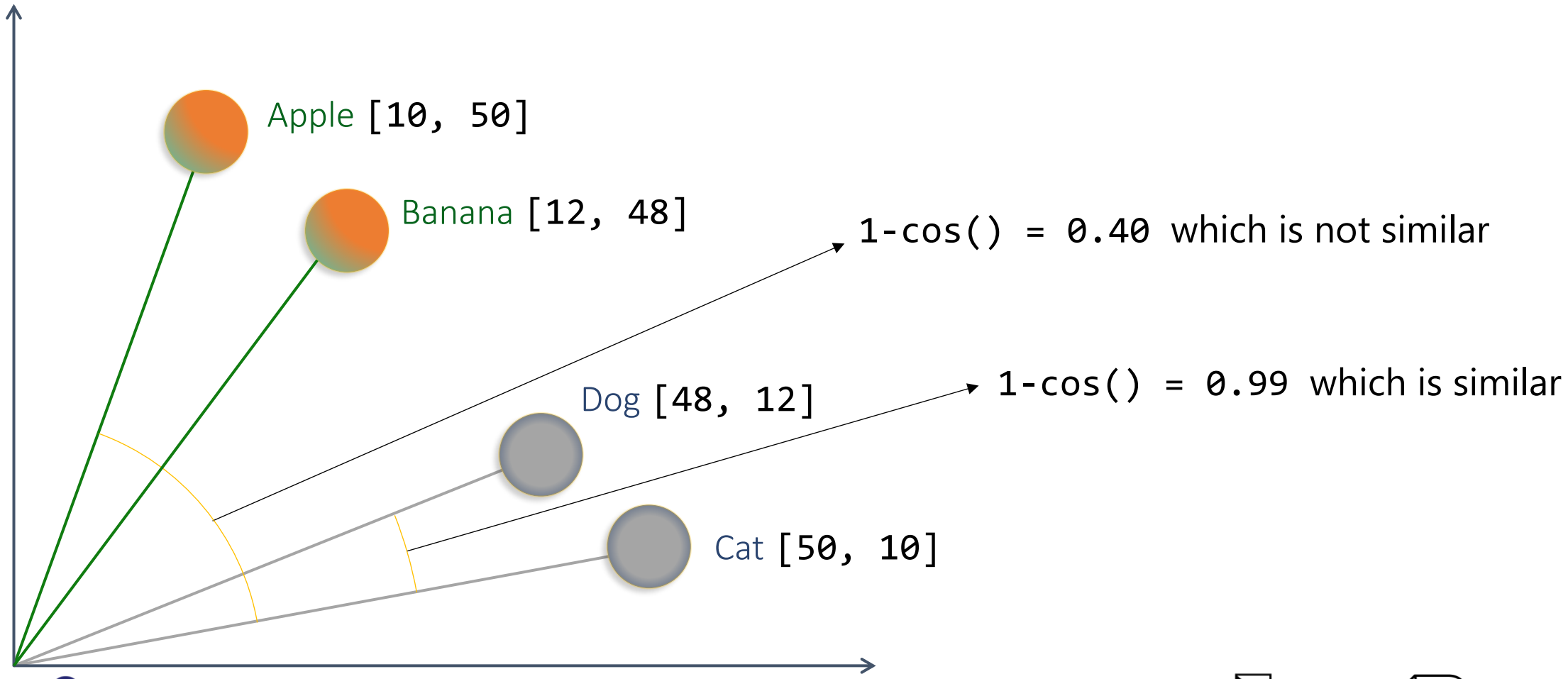
Vector generated by a model that has semantic meaning

```
dog: [0.9, 0.3, 0.2, ...]
Image of a dog: [0.9, 0.3, 0.2, ...]
puppy: [0.88, 0.33, 0.21, ...]

"I took the dog for a walk": [1.5, -0.8, 2.1, ...]
```

"dog"

"dog"


Apple [10, 50]
Banana [12, 48]
1-cos() = 0.40 which is not similar
Dog [48, 12]
1-cos() = 0.99 which is similar
Cat [50, 10]

VECTOR data type

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SQL Server 2025

Azure SQL

Managed Instances

SQL Server in Fabric

```
ALTER DATABASE SCOPED CONFIGURATION
SET PREVIEW_FEATURES = ON;
GO
```

column_name VECTOR({<dimensions>}) [NOT NULL | NULL]

A vector must have at least 1 dimension
The maximum number of dimensions supported is 1998

Each element of the vector is stored using single-precision (4 bytes) floating-point value (float32)

- Column-level constraints aren't supported, except for NULL/NOT NULL constraints.
    - DEFAULT and CHECK constraints aren't supported for **vector** columns.
    - Key constraints, such as PRIMARY KEY or FOREIGN KEY, aren't supported for **vector** columns. Equality, uniqueness, joins using vector columns as keys, and sort orders don't apply to **vector** data types.
    - There's no notion of uniqueness for vectors, so unique constraints aren't applicable.
    - Checking the range of values within a vector is also not applicable.
- Vectors don't support comparison, addition, subtraction, multiplication, division, concatenation, or any other mathematical, logical, and compound assignment operators.
- **vector** columns can't be used in memory-optimized tables.
- Altering **vector** columns using ALTER TABLE … ALTER COLUMN to other data types isn't permitted.

- Implicit and explicit conversion using CAST or CONVERT from the **vector** type can be done to **varchar**, **nvarchar** and **json** types. Similarly, only **varchar**, **nvarchar** and **json** can be implicitly or explicitly converted to the **vector** type.

- The vector type can't be used with the **sql_variant** type or assigned to a **sql_variant** variable or column. This restriction similar to **varchar(max)**, **varbinary(max)**, **nvarchar(max)**, **xml**, **json**, and CLR-based data types.

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VECTOR Functions

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Activate an LLM Model
Create a Master Key
Create HTTP Credential
Create external Model

All scripts here: lvercellati/DataSaturday80Pordenone

```sql
if not exists(select * from sys.database_scoped_credentials where [name] =
'https://my-azure-service.cognitiveservices.azure.com/')
begin
    create database scoped credential [https://my-azure-
service.cognitiveservices.azure.com/]
    with identity = 'HTTPEndpointHeaders', secret = '{"api-key":"..."}';
end
go
```

```sql
CREATE EXTERNAL MODEL MyEmbeddingModel
AUTHORIZATION dbo
WITH (
      LOCATION = 'https://my-azure-
service.cognitiveservices.azure.com/openai/deployments/text-embedding-3-
small/embeddings?api-version=2023-05-15',
      API_FORMAT = 'Azure OpenAI',
      MODEL_TYPE = EMBEDDINGS,
      MODEL = 'text-embedding-3-small',
      CREDENTIAL = [https://my-azure-service.cognitiveservices.azure.com/]
);
```

AI_GENERATE_EMBEDDINGS (source USE MODEL model_identifier)

```
AI_GENERATE_CHUNKS (source = text_expression
                    , chunk_type = 'FIXED'
                    [ [ , ] chunk_size = numeric_expression ]
                    [ [ , ] overlap = numeric_expression ]
                    )
```

Metric	Description	Range	Examples
`cosine`	Cosine (angular) distance	[0, 2]	`0`: identical vectors `2`: opposing vectors
`euclidean`	Euclidean distance	[-∞, +∞]	`0`: identical vectors
`dot`	Dot product-based indication of distance, obtained by calculating the *negative* dot product	[-∞, +∞]	Smaller numbers indicate more similar vectors

VECTOR_DISTANCE (distance_metric, vector1, vector2)

```
VECTOR_SEARCH(
    TABLE = object [AS source_table_alias]
    , COLUMN = vector_column
    , SIMILAR_TO = query_vector
    , METRIC = { 'cosine' | 'dot' | 'euclidean' }
    , TOP_N = k
) [AS result_table_alias]
```

```
CREATE VECTOR INDEX index_name
ON object ( vector_column )
[ WITH (
    [,] METRIC = { 'cosine' | 'dot' | 'euclidean' }
    [ [,] TYPE = 'DiskANN' ]
    [ [,] MAXDOP = max_degree_of_parallelism ]
    [ [,] DROP_EXISTING = { ON | OFF } ]
) ]
[ ON { filegroup_name | "default" } ]
[;]
```

K-Nearest Neighbors (KNN) is a traditional algorithm used to find the exact nearest neighbors of a query point in a dataset. While KNN is precise, it can be computationally expensive and slow, especially with large datasets.

Approximate Nearest Neighbors (ANN), on the other hand, aims to find neighbors that are close enough to the query point but not necessarily the exact nearest ones. ANN algorithms trade off a bit of accuracy for a significant gain in speed and efficiency, making them suitable for large-scale applications.

The current preview has the following limitations:

- Vector index can't be partitioned. **No partition support**.
- The table must have a **single column, integer, primary key clustered index**.
- Vector indexes aren't replicated to subscribers.
- A table with a vector index becomes **read only**. No data modification is allowed while the vector index is present on the table.

In Azure SQL Database and SQL database in Microsoft Fabric, you can set the ***ALLOW_STALE_VECTOR_INDEX*** *database scoped configuration to ON, which allows the table to be writable again.*

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GitHub - Azure-Samples/azure-sql-db-chatbot: Implement a RAG solution and call OpenAI right from Azure SQL DB to ask questions about your data

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Data Pied Piper

Power BI User Group Italy

Power BI User Group Italy - YouTube

THANKS!

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